

Power system:

Power Distribution Architecture:

Our Power distribution is relatively simple. The main voltages used will be 8.4-7.4v provided by our 2s LiPo battery, and 5v provided by our buck converter and our SBC/SBM.

7v4-8v4 → Motor driver, Buck converter.

5V → Raspberry pi & Nano, Sensors, Motor driver power, Servo motor.

3v3 → Cam (supplied by Pi/ribbon cable), May components (i.e All ToF, I2C multiplexer) are compatible with both 3v3 and 5V

Common ground will be the negative terminal of our LiPo pack to provide a common fixed voltage ref.

Power consumption analysis:

Assuming an unideal scenario:

Hardware	Voltage	Current draw
Raspberry pi zero W at 100% cpu capacity	5v	~220mA
Camera	5v	200-225mA
Arduino Nano (UART connection to PI)	3v3	30-50mA
All 6 VL53L1X (TBD)	5V-3v3	120mA (20mA per)
2 VL53L3CX (TBD)	3v3-5V	40mA (20mA per)
TB6612FNG driver	7.4v for motors, 5v consumption	5mA
TCA9548A	3v3-5V	~1mA, negligible
TOTAL	N/A	